# program 11

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## 11 . To gnenerate a basic box plot using ggplot2. enhanced with notches and outliers and grouped by a categorical variable using an in_built dataset in R.

### Step 1: Load the required library

```
library(ggplot2)
```

### Step 2: Explore the data set

```
data("iris")
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

```
str(iris)
```

```
'data.frame':   150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

### Step 3: Creating a box plot

```
ggplot(data = iris, aes(y = Sepal.Length ,x = Species))+
  geom_boxplot(
    outlier.alpha = NULL,
    notch = TRUE,
    staplewidth = 0.5,
    na.rm = FALSE,
    fill = "yellow",
    color = "black",
     outlier.color = "skyblue",
  outlier.fill = "red",
  orientation = NA,
```

```
show.legend = NA,
inherit.aes = TRUE
)+
labs(
  title = "Boxplot with notches and outliers",
  x = "Species",
  y = "Sepal.Legth"
)+
theme_minimal()
```

